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Tube end closure

Abstract

According to one aspect, a non-roll end sleeve is configured to be disposed on a seamless cylindrical paper tube, said non-roll sleeve having a rectangular cross-section perpendicular to a central axis of the cylindrical paper tube when disposed on the cylindrical paper tube. According to another aspect, a container includes a seamless paper tube having at least one end provided with a shape in cross-section generally having N sides, where N is an integer selected from 3 to 8, which at least one end is closed by flaps of material between openings cut into the at least one end that have been folded inwardly and secured in place to form an end closure.

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Background/Summary

BACKGROUND

(1) A cylindrical shape is conventionally employed in shipping containers for providing a desired structural integrity in the container, however, cylindrical containers often lack a practically stable upright orientation and also have a tendency to roll. As such, cylindrical tubes sometimes present difficulties when used as shipping containers due to their tendency to roll, especially when being transported on conveyor systems in sorting facilities. For example, rolling can cause the tubes to fall off the conveyor systems and/or cause the labels affixed to the tubes to not be properly positioned to be scanned by scanning devices associated therewith.

(2) Furthermore, cylindrical tubes often require end closures for retaining contents therein. A variety of end closures are known in the art for closing an open end of a tube, and non-limiting examples of end closures include wooden or plastic plugs, which can be inserted into the open end of the tube and stapled in place for retaining contents in the paper tube. Plugs can be inserted into and removed from an open end of a paper tube by hand, but can sometimes inadvertently disengage from the open end of the paper tube, which then allows the contents of the container to spill out of the container. For example, plugs can be caused to disengage from a tube end when contents of a tube shift in the tube and strike an interior side of the plug, or when the tube is compressed and an internal pressure of the tube is increased.

BRIEF DESCRIPTION

(3) According to one aspect, a non-roll end sleeve is configured to be disposed on a seamless cylindrical paper tube, said non-roll sleeve having a rectangular cross-section perpendicular to a central axis of the cylindrical paper tube when disposed on the cylindrical paper tube.

(4) According to another aspect, a container includes a seamless paper tube having at least one end provided with a shape in cross-section generally having N sides, where N is an integer selected from 3 to 8, which at least one end is closed by flaps of material between openings cut into the at least one end that have been folded inwardly and secured in place to form an end closure.

(5) According to another aspect, a method of making a container includes compressing at least one open end of a seamless paper tube having a circular cross-section to have a polygonal shape with N sides, where N is an integer selected from the group consisting of 3, 4, 5, 6, 7 and 8. The method also includes cutting a plurality of openings into the compressed open end of the seamless paper tube, and folding material between the plurality of openings inwardly to form flaps. The method also includes overlapping at least two of said flaps to close off the compressed open end of the seamless paper tube, and securing the flaps in place to form a first end closure.

(6) According to another aspect, a container includes a seamless paper tube having at least one end provided with a shape in cross-section generally having N sides, where N is an integer selected from 3 to 8, which at least one end is closed by flaps of material between openings cut into the at least one end that have been folded inwardly and secured in place to form an end closure, and a non-roll extension protruding from said end closure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partial perspective view of a non-roll end sleeve assembled on a tube.
- (2) FIG. 2 is a front view of the non-roll end sleeve in a disassembled condition.
- (3) FIG. 3 is a partial perspective view of a non-roll end sleeve according to another aspect, assembled on a tube.
- (4) FIG. 4 is a partial perspective view of the non-roll end sleeve of FIG. 3.
- (5) FIG. 5 is a partial perspective view of the non-roll end sleeve of FIG. 3.
- (6) FIG. 6 is a partial perspective view of a non-roll end sleeve according to another aspect, assembled on a tube.
- (7) FIG. 7 is a partial perspective view of a non-roll end sleeve according to another aspect.
- (8) FIG. 8 is a partial perspective view of the non-roll end sleeve of FIG. 7, assembled on a tube.
- (9) FIG. 9 is a partial perspective view of a non-roll end sleeve according to another aspect, assembled on a tube.
- (10) FIG. 10 is a partial perspective view of a tube according to another aspect.
- (11) FIG. 11 is a partial perspective view of the tube of FIG. 10.
- (12) FIG. 12 is a top view of the tube of FIG. 10.
- (13) FIG. 13 is a top view of the tube of FIG. 10.
- (14) FIG. 14 is a top view of the tube of FIG. 10.
- (15) FIG. 15 is a top view of the tube of FIG. 10.
- (16) FIG. 16 is a partial side view of a tube according to another aspect.
- (17) FIG. 17 is a perspective view of the tube of FIG. 10.
- (18) FIG. 18 is a partial perspective view of a tube according to another aspect.
- (19) FIG. 19 is a partial perspective view of a tube according to another aspect.
- (20) FIG. 20 is a partial perspective view of the tube of FIG. 19.
- (21) FIG. 21 is a partial perspective view of a tube according to another aspect.
- (22) FIG. 22 is a partial perspective view of the tube of FIG. 21.
- (23) FIG. 23 is a partial perspective view of the tube of FIG. 21.
- (24) FIG. 24 is a top view of the tube of FIG. 21.
- (25) FIG. 25 is a partial perspective view of a tube according to another aspect.
- (26) FIG. 26 is a partial perspective view of the tube of FIG. 25.
- (27) FIG. 27 is a partial perspective view of a tube according to another aspect.
- (28) FIG. 28 is a partial perspective view of a tube according to another aspect.
- (29) FIG. 29 is a perspective view of a tube according to another aspect.
- (30) FIG. 30 is a partial perspective view of a tube according to another aspect.

DETAILED DESCRIPTION

(31) It should, of course, be understood that the description and drawings herein are merely illustrative and that various modifications and changes can be made in the structures disclosed without departing from the present disclosure. Referring now to the drawings, wherein like numerals refer to like parts throughout the several views, FIG. 1 depicts a non-roll end sleeve **100** assembled on a tube **102**. The tube **102** is received in a sleeve opening **104** of the non-roll end sleeve **100** at a partially closed end **110** of the non-roll end sleeve **100**. The end is partially closed because the flap **150** that closes the non-roll end sleeve **100** includes a sleeve opening **104**. The sleeve opening **104** is configured to close around the tube **102** so as to fix the non-roll end sleeve **100** with the tube **102** at the partially closed end **110**. An end closure **112** as described in Applicant's U.S. Pat. No. 7,296,730 B2 is inserted into a tube opening **114** at an end **120** of the tube **102** to close off the tube opening **114**. In an embodiment, the tube **102** is a seamless cylindrical tube having a circular profile along a central axis **122** of the tube **102**, where the term "seamless" as used herein means that the tube **102** does not have a seam or a joint that runs parallel to the central axis **122** or otherwise along a longitudinal direction of the tube **102**, and in this manner a seamless

tube is different than a container that is formed by, for example, rolling or folding a piece of paperboard or paper-based material to form a hollow structure which is then taped or otherwise secured so as to have such an elongate seam.

(32) As shown, an outer perimeter **129** of the end closure **112** is disposed on a front face **130** of the non-roll end sleeve **100** in which the sleeve opening **104** is defined, along an inner perimeter **132** of the sleeve opening **104**. The end closure **112** is configured to abut the non-roll end sleeve **100** on the tube **102** in a longitudinal direction of the tube **102**, and in this manner the end closure **112** is configured to retain the non-roll end sleeve **100** on the tube **102**.

(33) Rectangular edges **134** of the partially closed end **110** of the non-roll end sleeve **100** are configured to support the tube **102** on a supporting surface (not shown), such as a belt in a conveyor system. The non-roll end sleeve **100** is tapered inward toward the tube **102** along a longitudinal direction of the tube **102** from the partially closed end **110** to an opposite end **140** of the non-roll end sleeve **100** in a longitudinal direction of the tube **102**. With this construction, when the tube **102** with the non-roll end sleeve **100** is positioned on a substantially horizontal surface and oriented substantially horizontally, the rectangular edges **134** of the partially closed end **110** of the non-roll end sleeve **100** are configured to rest on the substantially horizontal surface, with sides **142** of the non-roll end sleeve **100** spaced from the substantially horizontal surface.

(34) The front face **130** of the non-roll end sleeve **100** includes a tab **144**, shown in hidden lines, extended from the front face **130** and bent toward the tube **102**. The tab **144** is folded under the sides **142** of the non-roll end sleeve **100** and fixed thereto with an adhesive for supporting the front face **130** of the non-roll end sleeve **100** on the sides **142** of the non-roll end sleeve **100**.

(35) FIG. 2 depicts a front view of the non-roll end sleeve **100** disassembled from a tube, in an unfolded configuration. The non-roll end sleeve **100** includes a flap **150** configured to form the front face **130** when the non-roll end sleeve **100** is in a folded configuration. The flap **150** is shaped as a square and includes the tab **144** extended therefrom. While the flap **150** is depicted as square shaped, the flap **150** may alternatively be rectangular or otherwise shaped so as to partially close the non-roll end sleeve **100** around the tube **102** without departing from the scope of the present disclosure.

(36) The non-roll end sleeve **100** also includes five panels **152** configured to form the sides **142** of the non-roll end sleeve **100** when the non-roll end sleeve **100** is in the folded configuration. The panels **152** are respectively shaped as trapezoids and organized relative to each other to form a tapered structure in the non-roll end sleeve **100** when the panels **152** are folded together in the folded configuration as in FIG. 1. More specifically, with continued reference to FIG. 2, a first panel **154** is continuous with and configured for being folded relative to a second panel **160**, the second panel **160** is continuous with and configured for being folded relative to a third panel **162**, the third panel **162** is continuous and configured for being folded relative to a fourth panel **164**, and the fourth panel **164** is continuous with and configured for being folded relative to a fifth panel **170** such that the first panel **154** and the fifth panel **170** are in contact with each other and overlapping along the longitudinal direction of the tube **102** when the non-roll end sleeve **100** is in the folded configuration and assembled with the tube **102**. When the non-roll end sleeve **100** is assembled with the tube **102** as in FIG. 1, the first panel **154** and the fifth panel **170** are fixed to each other with adhesive.

(37) In an embodiment, a non-roll end sleeve is not tapered. To this end, FIG. 3 depicts a partial perspective view of a non-roll end sleeve **200** having sides **202** that are not tapered along the longitudinal direction of the tube **102**. As such, an outer surface **204** of the non-roll end sleeve **200** in any cross-section taken through the non-roll end sleeve **200** perpendicular to a central axis **210** of the tube **102** would define a rectangle having a size and shape consistent with any other cross-section along the non-roll end sleeve **200** perpendicular to the central axis **210**. With this construction, when the tube **102** with the non-roll end sleeve **200** is positioned on a substantially horizontal surface and oriented substantially horizontally, rectangular edges **212** of a partially

closed end **214** of the non-roll end sleeve **200** and the sides **202** of the non-roll end sleeve **200** are configured to rest on the substantially horizontal surface.

(38) In an embodiment, an inwardly facing tab (not shown) is formed in a sleeve opening **220** at the partially closed end **214** of the non-roll end sleeve **200**. When the non-roll end sleeve **200** is assembled with the tube **102**, the tube **102** is received in the sleeve opening **220** and the tab is pressed into a slot or notch (not shown) cut into the end **120** of the tube **102**, and is further pressed against an inner side **222** of the tube **102** when the end closure **112** is inserted into the tube opening **114** to close off the tube opening **114**. With this construction, the end closure **112** presses the tab from the non-roll end sleeve **200** against the inner side **222** of the tube **102** and prevent the non-roll end sleeve **200** from being displaced from a desired location on the tube **102**. In an alternative embodiment, the tab can be folded over an edge **224** of the tube **102**.

(39) FIG. 4 depicts a partial perspective view of the non-roll end sleeve **200** where, as shown, each side **202** of the non-roll end sleeve **200** respectively includes flaps **230** respectively disposed in an unbent position with respect to the non-roll end sleeve **200** such that each flap **230** is extended from the non-roll end sleeve **200** in the longitudinal direction of the tube **102**, away from the end **120** of the tube **102**. With each flap **230** respectively in the unbent position, an open end **232** of the non-roll end sleeve **200** is configured for receiving the tube **102** without a flap **230** contacting the tube **102**.

(40) FIG. 5 depicts a partial perspective view of the non-roll end sleeve **200** where each flap **230** is respectively bent inward from the corresponding side **202** of the non-roll end sleeve **200** into a bent position directed toward the central axis **122** of the tube **102**. As shown, each flap **230** contacts an outer surface **234** of the tube **102** so as to fix the open end **232** of the non-roll end sleeve **200** with the tube **102**, thereby supporting the tube **102** in the non-roll end sleeve **200** at the open end **232**. Unless otherwise stated, the non-roll end sleeve **200** functions in a similar manner and has similar features as the non-roll end sleeve **100**.

(41) In an embodiment, a non-roll end sleeve is configured for assembly on a tube, spaced from the end of the tube. To this end, FIG. 6 depicts a partial perspective view of a non-roll end sleeve **300** assembled on the tube **102** where the non-roll end sleeve **300** is spaced from the end **120** of the tube **102** in the longitudinal direction of the tube **102**, and spaced from the end closure in the longitudinal direction of the tube **102**. Unless otherwise stated, the non-roll end sleeve **300** functions in a similar manner and has similar features as the non-roll end sleeve **100**.

(42) In an embodiment, a non-roll end sleeve includes a front face configured to close over a tube opening. To this end, FIG. 7 depicts a partial perspective view of a non-roll end sleeve **400** including a first flap **402** and a second flap **404** as a pair of side flaps configured for being folded toward each other, over the end **120** of the tube **102** in the longitudinal direction of the tube **102**. The non-roll end sleeve **400** also includes a third flap **410** as a cover configured for being folded over the first flap **402** and the second flap **404** in the longitudinal direction of the tube **102**, on a side of the first flap **402** and the second flap **404** opposite from the end **120** of the tube **102**. The third flap **410** includes a third flap tab **412** extended therefrom and configured for being folded from the third flap **410** and inserted between the first flap **402**, the second flap **404**, and a side **414** of the non-roll end sleeve **400** opposite from the third flap **410** such that the third flap tab **412** retains the third flap **410** folded over the first flap **402**, the second flap **404**, and the end **120** of the tube **102** when the non-roll end sleeve **400** is assembled with the tube **102**. The third flap **410** also includes a third flap slot **420** defined therethrough, along a side from which the third flap **410** extends.

(43) FIG. 8 depicts a partial perspective view of the non-roll end sleeve **400** assembled on the tube **102**. As shown, the third flap **410** is folded over the end **120** of the tube **102** and forms a closed end **422** of the non-roll end sleeve **400** over the end **120** of the tube **102** in the longitudinal direction of the tube **102**. An opposite end **424** of the non-roll end sleeve **400** is located across sides **430** of the non-roll end sleeve **400** in the longitudinal direction of the tube **102**.

(44) The sides **430** of the non-roll end sleeve **400** include a bottom side **432** of the non-roll end sleeve **400** positioned on a supporting surface (not shown) for the non-roll end sleeve **400**, such as a top surface of a conveyor belt or storage shelf. The sides **430** of the non-roll end sleeve **400** also include a top side **434** of the non-roll end sleeve **400** defining a side of the non-roll end sleeve **400** opposite from the bottom side **432**. The top side **434** of the non-roll end sleeve **400** includes a predefined location for a label **440** corresponding with the tube **102**. When the bottom side **432** of the non-roll end sleeve **400** supporting the non-roll end sleeve **400** in an upright position, the label **440** on the top side **434** of the non-roll end sleeve **400** is visible from a top view of the non-roll end sleeve **400**.

(45) In an embodiment, a tube flap extended from a tube extends through a closed end of a non-roll sleeve. To this end, FIG. **9** depicts a partial perspective view of the non-roll end sleeve **400** disposed on a tube **442** with a tube flap **444** received in and disposed through the third flap slot **420**. The third flap slot **420** is configured for receiving the tube flap **444** when the third flap **410** is folded over an end **450** of the tube **442** and in this manner retaining the third flap **410** and the non-roll end sleeve **400** on the tube **442**. With the tube flap **444** extended through the third flap slot **420**, the tube flap **444** retains the non-roll end sleeve **400** on the tube **442** in a longitudinal direction of the tube **442**, and fixes a radial position of the tube **442** with respect to the non-roll end sleeve **400**. Unless otherwise stated, the non-roll end sleeve **400** and the tube **442** respectively function in a similar manner and have similar features as the non-roll end sleeve **100** and the tube **102**.

(46) In an embodiment, a tube is shaped to have a substantially square shaped profile. To this end, FIG. **10** depicts a partial perspective view of a tube **500** disposed in a press **502** with an end **504** of the tube **500** pressed between four rollers **510** such that the tube **500** is shaped to have a substantially square shaped profile along a longitudinal direction of the tube **500** at the end of the tube **500**. It will be appreciated that other forming methods can be used (e.g., sliding the paper tube on a former and rolling/pressing the tube onto the former to define the corners of the profile).

(47) In a particularly preferred method of shaping a paper tube employed as the tube **500**, and the paper tube is formed with a sufficiently high moisture content such that the tube **500** is pliable for being shaped and subsequently dried having a square profile along the longitudinal direction of the tube **500**. In an embodiment, the paper used to form the tube has a 5-6% moisture content and an adhesive in the tube **500** has a relatively high water content such that when the tube **500** is initially manufactured, the tube **500** has a moisture content of about 20% to enable shaping the tube **500** with a square profile along the longitudinal direction of the tube **500**. In a dried condition where the tube **500** is sufficiently rigid and has sufficient structural integrity so as to reasonably maintain its shape as a shipping container, the tube **500** has a moisture content of about 8%. It will be appreciated that forming can be performed when the tubes are in a dried condition, if desired. But results obtained when forming is performed on “dry” tubes are generally not as optimal as when forming is performed on “wet” tubes.

(48) While the tube **500** is depicted as being pressed and formed into a square profile, rollers **510** in the press **502** can be otherwise organized to compress the tube **500** into a rectangle having sides that in sum are equal to a circumference of the seamless paper tube without departing from the scope of the present disclosure. Notably, because a circumference of the tube **500** remains constant when pressed into a square profile, predicting a length of sides **512** of the tube **500** in a square profile at the end **504**, based on the size of the tube **500** and the tube having a circular profile before being pressed, is possible by measuring a diameter of the tube **500**, calculating a corresponding circumference of the tube **500**, and dividing the calculated circumference by four. Also, a distance between opposite corners **520** of the tube **500** can be calculated by taking a square root of a sum of squares of two consecutive sides.

(49) Further, the number of rollers **510** employed in pressing the tube **500** can be adjusted to compress the tube **500** into a variety of profiles including but not limited to polygonal profiles without departing from the scope of the present disclosure. For example, three rollers **510** can be

arranged to press the tube **500** into a triangular profile, and alternatively six rollers **510** can be arranged to press the tube **500** into a hexangular profile. In this manner, the end **504** of the tube **500** can be pressed to a profile having N sides **512** formed from N rollers **510**, where N is an integer greater than 2. In an embodiment, N is specifically an integer selected from a range from 3 to 8. In a further embodiment, N is 4, 5, 6, 7, or 8.

(50) FIG. **11** depicts a partial perspective view of the tube **500**, which has been squared at the end **504** and has slits **514** cut at squared corners **520** of the end **504** to form four flaps **522** configured for being folded inward, into the square profile of the end **504** of the tube **500**. The slits **514** have a length that is about half of a circumference of the tube **500**.

(51) FIGS. **12-15** depict the end **504** of the tube **500** folded into an end closure **524**. As shown in FIG. **12**, a portion of the tube **500** between two consecutive slits **514** taken along a circumference of the tube **500** forms a first flap **530** and is folded inward. The first flap **530** spans approximately half of the diameter of a square formed by compressing the tube **500** into a square. With this, as shown in FIG. **13**, a distal edge **532** of an opposing second flap **534** formed between two consecutive slits **514**, folded inward toward the first flap **530**, meets a distal edge **540** of the first flap **530**. FIG. **14** depicts a third flap **542** folded inward with respect to the tube **500** over the first flap **530** and the second flap **534** in the longitudinal direction of the tube **500**. The third flap **542** spans approximately half of the diameter of a square formed by compressing the tube **500** into a square and when folded over the first flap **530** and the second flap **534**, and covers half of the first flap **530** and half of the second flap **534** in the longitudinal direction of the tube **500**. As shown in FIG. **15**, a fourth flap **544** is folded inward with respect to the tube **500**, over the second flap **534** and the third flap **542** in the longitudinal direction of the tube **500**. The fourth flap **544** spans approximately half of the diameter of a square formed by compressing the tube **500** into a square and when folded over the first flap **530** and the second flap **534** such that a distal edge **550** of the third flap **542** meets a distal edge **552** of the fourth flap **544** when the third flap **542** and the fourth flap **544** are folded inward toward each other with respect to the tube **500**. As such, the fourth flap **544** covers half of the second flap **534** and half of the third flap **542** in the longitudinal direction of the tube **500**.

(52) Adhesive tape **554** shown in hidden lines is employed to fix the third flap **542** and the fourth flap **544** folded over the first flap **530** and the second flap **534**. Notably, adhesive and staples may be employed in combination with or alternative to the adhesive tape **554** in fixing the first flap **530**, the second flap **534**, the third flap **542**, and the fourth flap **544** folded inward with respect to the tube **500** without departing from the scope of the present disclosure. In an embodiment, adhesive is disposed on an outside of the first flap **530**, an outside of the second flap **534**, an inside of the third flap **542**, and an inside of the fourth flap **544** in the longitudinal direction of the tube **500**, and the adhesive tape **554** is a release tape disposed on an outside of the third flap **542** and the fourth flap **544** while the adhesive fixes the inside of the third flap **542** and the inside of the fourth flap **544** with the outside of the first flap **530** and the outside of the second flap **534**. The adhesive tape **554** can be removed from the third flap **542** and the fourth flap **544** when the adhesive fixing the first flap **530**, the second flap **534**, the third flap **542**, and the fourth flap **544** folded inward with respect to the tube **500** dries.

(53) In an embodiment, a tube end closure is folded with a flap extending outward therefrom in a longitudinal direction of the tube. To this end, FIG. **16** depicts a partial side view of a tube **600** having an end closure **602** formed from a first flap **604**, and having a second flap **610** extending from the end closure **602** in a longitudinal direction of the tube **600**. As shown in FIG. **17**, an end **612** of the tube **600** is pressed and folded into the end closure **602**, and an opposite end **614** of the tube **600** is similarly pressed and folded into an opposite end closure **620**. Unless otherwise stated, the opposite end **614** of the tube **600** functions in a similar manner and has features similar to the end **612** of the tube **600**. The end **612** of the tube **600** and the opposite end **614** of the tube **600** are pressed and folded such that the tube **600** has a circular profile between the end **612** of the tube **600**

and the opposite end **614** of the tube **600** along the longitudinal direction of the tube **600**. Unless otherwise stated, the tube **600** functions in a similar manner and has similar features as the tube **102**.

(54) In an embodiment, a tube is shaped to have a polygonal profile along a longitudinal direction of the tube, along a length of the tube from an end of the tube to an opposite end of the tube. To this end, FIG. **18** depicts a partial perspective view of a tube **700** having a rectangular profile from an end **702** of the tube **700** forming an end closure **704** to an opposite end **710** of the tube **700** forming an opposite end closure **712**. The substantially rectangular profile of the tube **700** includes squared corners **714** disposed along the longitudinal direction of the tube **700**, the corners **714** being interposed between and separating sides **720** of the tube **700**. Unless otherwise stated, the tube **700** functions in a similar manner and has similar features as the tube **102**.

(55) In an embodiment, a tube is shaped to have a triangular profile along a longitudinal direction of the tube. To this end, FIG. **19** depicts a partial perspective view of a tube **800** shaped to have a triangular profile including three sides **802** elongated in a longitudinal direction of the tube **800** and respectively interposed between and separating three corners **804** at an end of the tube **800**.

(56) FIG. **20** is a partial perspective view of the tube **800** having slits **810** cut into the end **812** of the tube **800** respectively along the corners **804** of the tube **800**. The slits **810** are interposed between and separate the sides **802** of the tube **800** at the end **812** of the tube **800** such that the elongated surfaces respectively form a first flap **814**, a second flap **820**, and a third flap **822** located at the end **812** of the tube **800** and configured for being folded inward with respect to the tube **800**.

(57) The first flap **814**, the second flap **820**, and the third flap **822** are configured for folding inward with respect to the tube **800** and overlap each other to form an end closure. To this end, the first flap **814**, the second flap **820**, and the third flap **822** are configured for being folded or cut along dashed lines **830** which respectively delineate a triangular area **832** of the corresponding flap **814**, **820**, **822**, the triangular area being configured for covering the triangular profile of the tube **800** in the longitudinal direction of the tube **800**.

(58) Notably, by respectively removing material from the first flap **814**, the second flap **820**, and the third flap **822** to form a triangular flap corresponding with the triangular profile of the tube **800**, the first flap **814**, the second flap **820**, and the third flap **822** are respectively configured for being folded inward with respect to the tube **800** to form an end closure without extending beyond the triangular profile of the tube **800** in a radial direction of the tube **800**. In an embodiment, the second flap **820** is cut into a triangle, and the first flap **814** and the third flap **822** are folded respectively along the corresponding dashed lines **830** to form a triangular area, where folded material of the first flap **814** is inserted into the end **812** of the tube **800** and folded material from the third flap **822** is folded around an outside surface **834** of the tube **800** at the end **812**. Unless otherwise stated, the tube **800** functions in a similar manner and has similar features as the tube **102**.

(59) In an embodiment, a tube is shaped to have a hexangular profile along a longitudinal direction of the tube. To this end, FIGS. **21-24** depict a tube **900** forming an end closure **902** in a hexangular shape. As shown in FIG. **21**, the tube **900** has a hexangular profile including six sides **904** elongated in a longitudinal direction of the tube **900**, the sides **904** respectively interposed between and separating six corners **910** at an end of the tube **900**. Slits **912** cut into an end **914** of the tube **900** along the corners **910** respectively form flaps **920** from the sides **904** located between consecutive slits **912**. The flaps **920** are configured for being folded inward with respect to the tube **900**, in a radial direction of the tube **900** so as to form the end closure **902**. In an embodiment, the end **914** of the tube **900** is not formed to have a hexangular profile before the flaps **920** are folded inward to form the end closure **902**. FIG. **22** depicts the tube **900** where three consecutive flaps **920** are folded inward, partially forming the end closure **902** at the end **914** of the tube **900**. FIG. **23** depicts the tube **900** where the flaps **920** are folded inward, in the radial direction of the tube **900**, forming the end closure **902**. As shown in FIG. **24**, the flaps **920** are fixed in the end closure **902** by adhesive tape **922**. Notably, the flaps **920** may be fixed in the end closure **902** with adhesive or

staples in combination with or alternative to the adhesive tape **922** without departing from the scope of the present disclosure. Unless otherwise stated, the tube **900** functions in a similar manner and has similar features as the tube **102**.

(60) In an embodiment, a tube includes a flap that is sufficiently elongated so as to extend outward beyond an end closure of the tube in a radial direction of the tube. To this end, FIG. 25 depicts a partial perspective view of a tube **1000** formed including four flaps **1002** at an end **1004** of the tube **1000**. The flaps **1002** are configured for being folded with respect to the tube **1000**, at the end **1004** of the tube **1000**. One of the flaps **1002** is an elongated flap **1010** having a length greater than a diameter of the tube **1000**. With this construction, as shown in FIG. 26, the elongated flap **1010** extends in a radial direction of the tube **1000**, beyond an end closure **1012** formed by the flaps **1002** when the elongated flap **1010** is folded into the end closure **1010**. By extending beyond the end closure **1012** in the radial direction of the tube **1000**, the elongated flap **1010** forms a non-roll extension in the radial direction of the tube **1000**. The flaps **1002**, including the elongated flap **1010** as the non-roll extension, are formed integrally with the tube **1000**. Unless otherwise stated, the tube **1000** functions in a similar manner and has similar features as the tube **102**.

(61) In an embodiment, a non-roll extension is inserted in an end closure of a tube and extends from multiple sides of the tube. To this end, FIG. 27 depicts a partial perspective view of a tube **1100** including an end closure **1102** with a non-roll extension **1104** inserted therein. The non-roll extension **1104** is a strip of corrugated fiberboard placed at an end **1110** of the tube **1100**, inserted in the end closure **1102**, and fixed in the end closure **1010** with adhesive tape **1112** such that a first end **1114** and a second end **1120** of the non-roll extension **1104** respectively extend from opposing sides of the end **1110** of the tube **1100**. As depicted, the non-roll extension **1104** is fixed between an inner flap **1122** and an outer flap **1124** in the end closure **1102**, however the non-roll extension **1104** can be fixed on an outside surface **1130** of the outer flap **1124** without departing from the scope of the present disclosure. The non-roll extension **1104** is flexible in a longitudinal direction of the tube **1100** and relatively rigid in a direction perpendicular to the longitudinal direction of the tube **1100** so as to inhibit the tube **1100** from rolling about a central axis **1132** of the tube **1100**, but being able to bend inward in the longitudinal direction of the tube **1100** (i.e., toward the tube) for a relatively compact storage configuration of the tube (the non-roll extension would spring outwardly into the non-roll configuration when not in the bent/inwardly biased storage configuration).

(62) The non-roll extension **1104** is formed from a material sufficiently elastic to bend in the longitudinal direction of the tube **1100** and maintain a spring bias toward a radial direction of the tube **1100** so as to automatically return to an extended position with respect to the tube **1100** when not restricted in the storage configuration. While the non-roll extension **1104** is disclosed as being formed from fiberboard, the non-roll extension **1104** may alternatively be formed from a material with sufficient elasticity to similarly exert a spring bias in the radial direction of the tube **1100**, such as a polymer material. Unless otherwise stated, the tube **1100** functions in a similar manner and has similar features as the tube **102**.

(63) In an embodiment, a non-roll extension is inserted in an end closure of a tube and extends from a side of the tube. To this end, FIG. 28 depicts a partial perspective view of a tube **1200** including an end closure **1202** with a non-roll extension **1204** inserted therein. The non-roll extension **1204** is a strip of corrugated fiberboard placed at an end **1210** of the tube **1200**, inserted in the end closure **1202**, and fixed in the end closure **1202** with adhesive tape **1212** such that a first end **1214** of the non-roll extension **1204** extends from a side of the tube **1200** and a second end **1220** of the non-roll extension **1204** is inserted in the end closure **1202**. As shown in FIG. 29, an opposite end **1222** of the tube **1200** similarly includes an opposite non-roll extension **1224** inserted therein at an opposite end closure **1230**. Unless otherwise stated, the opposite end **1222** of the tube **1200**, the opposite non-roll extension **1224**, and the opposite end closure **1230** respectively function in a similar manner and have similar features as the end **1210** of the tube **1200**, the non-roll extension **1204**, and the end closure **1202**. Also, unless otherwise stated, the tube **1200**

functions in a similar manner and has similar features as the tube **102**.

(64) In an embodiment, a non-roll extension extending from an end closure of a tube is folded back and fixed to a side of the tube. To this end, FIG. **30** depicts a partial perspective view of a tube **1300** with a non-roll extension **1302** inserted in an end closure **1304** at an end **1310** of the tube **1300**. The non-roll extension **1302** extends radially outward from the end closure **1304** with respect to the tube **1300**, and is bent back toward the tube **1300** along a longitudinal direction of the tube **1300** with a distal end **1312** of the non-roll extension **1302** fixed to an exterior surface **1314** of the tube **1300** at a side of the tube **1300**. The non-roll extension **1302** is bent to form a relatively rigid triangular shape. As depicted, the non-roll extension **1302** extends from one side of the tube **1300**, however, the non-roll extension may extend from multiple side of the tube **1300** without departing from the scope of the present disclosure. Unless otherwise stated, the tube **1300** functions in a similar manner and has similar features as the tube **102**.

(65) It will be appreciated that various embodiments of the above-disclosed and other features and functions, or alternatives or varieties thereof, may be desirably combined into many other different systems or applications. Also that various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

Claims

1. A shipping container comprising: a seamless cylindrical paper tube having a central axis and first and second ends each defining an opening; an end closure that is received within and closes the opening defined by one of the first and second ends of the tube; and a non-roll sleeve disposed on the tube at the end of the tube defining the opening within which the end closure is received; wherein the non-roll sleeve has a rectangular cross-section perpendicular to the central axis of the tube when disposed on the tube, and wherein the non-roll sleeve disposed on the tube covers only a portion but not all of the tube, leaving a central portion of the tube exposed.
2. The shipping container according to claim 1, wherein the non-roll sleeve disposed on the tube narrows along its length in a direction away from the end of the tube.
3. The shipping container according to claim 1, wherein the non-roll sleeve has the same cross-section at any point along the central axis of the tube when disposed on the tube.
4. The shipping container according to claim 1, wherein the end closure secures the non-roll sleeve on the tube.
5. The shipping container according to claim 1, wherein the rectangular cross-section is square.
6. A shipping container comprising a seamless paper tube having an exposed central portion with a circular cross-section and first and second ends, wherein at least one of the first and second ends has a non-circular, polygonal cross-section, wherein the polygonal cross-section has N sides, wherein N is an integer selected from 3 to 8, wherein a plurality of foldable flaps are provided at the end with the non-circular, polygonal cross-section, and wherein the end with the non-circular, polygonal cross-section is closable by folding the flaps inwardly and securing the flaps in place to form an integral end closure.
7. The shipping container according to claim 6, wherein at least two of said flaps are configured to overlap each other when folded inwardly.
8. The shipping container according to claim 6, wherein at least two of said flaps are configured to have abutting edges when folded inwardly.
9. The shipping container according to claim 6, wherein both ends of the tube are non-circular and polygonal in cross-section.
10. The shipping container according to claim 6, further comprising a non-roll extension configured to protrude from the end closure.
11. The shipping container according to claim 10, wherein the non-roll extension is a separate strip

of material configured to be inserted into the end closure.

12. The shipping container according to claim 10, wherein the non-roll extension is a portion of a flap of material configured to be folded inwardly to form the end closure.

13. The shipping container according to claim 10, wherein the non-roll extension is flexible.

14. The shipping container according to claim 10, wherein an end of the non-roll extension is configured to be affixed to a side of the tube inward of the end closure.

15. The container according to claim 14, wherein the non-roll extension includes a fold and is triangular in cross-section when affixed to a side of the tube inward of the end closure.

16. A method comprising: placing matter into a shipping container according to claim 6; folding the flaps inwardly; and securing the flaps in place to form the end closure.

17. The method according to claim 16, wherein the flaps are secured in place using adhesive tape.
